Computer Architecture Laboratory

Milestone 1: Warm-up

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1 Overview

Block	Tested behavior	Result
Adder	Add/subtract; signed/unsigned; carry/borrow; zero and signed extrema.	PASS
Magnitude Comparator	Equality; signed/unsigned ordering; sign boundaries; one-hot outputs.	PASS
Shifter	SRL/SLL/SRA; amounts 0-31; zero/sign fill; reserved mode.	PASS
Arithmetic Logic Unit	Ten operations; unused encodings; SLT/SLTU; shift-amount masking.	PASS
Register File	All register addresses; x0; edge write; dual asynchronous reads; reset; write disable.	PASS
Asynchronous-read Memory	All byte masks and offsets; little-endian lanes; valid/invalid addresses; readback.	PASS
Synchronous-read Memory (Optional)	Enabled reads; disabled-read hold; read-before-write; minimum size and no address wrap.	PASS

2 Adder

WF01 | Basic arithmetic

Item	Testcase details
Input	a/b = 0/1 and 1/0; sub=0/1; signed/unsigned modes.
Output	0+1=1; 0-1=FFFFFFFF; 1-0=1.
Test	Basic addition, subtraction and zero identity.
Result	PASS

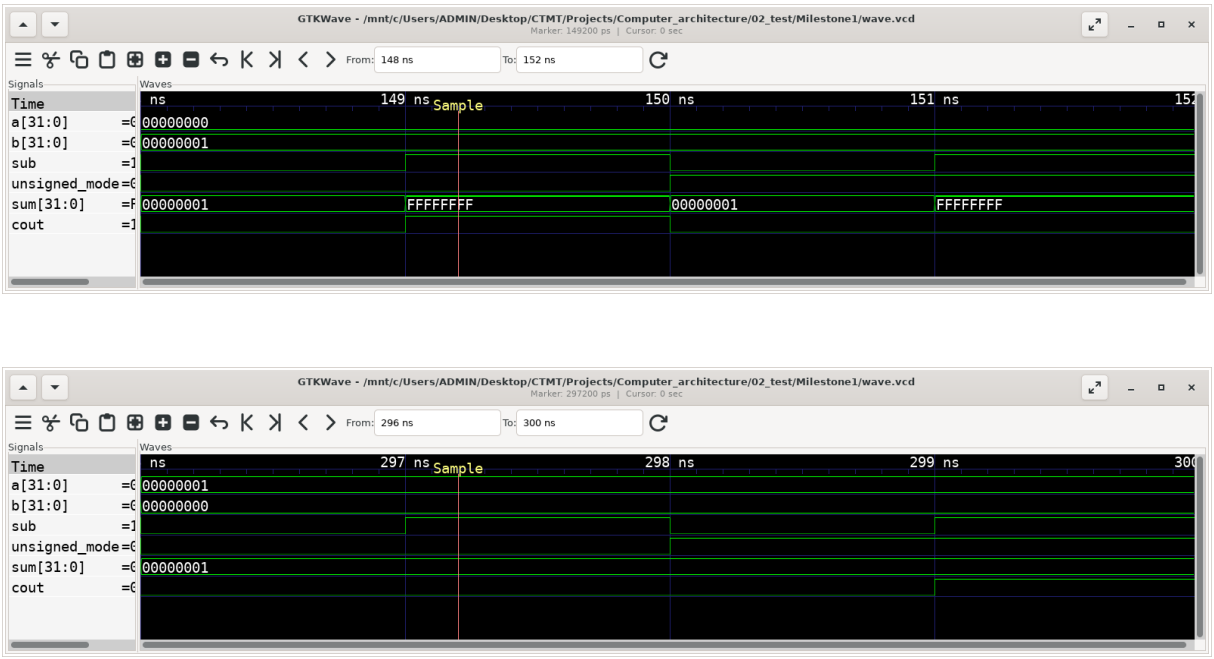


Figure WF01. Basic arithmetic.

WF02 | Carry / borrow

Item	Testcase details
Input	a/b = FFFFFFFF/1 and 0/1; unsigned mode; add/sub.
Output	FFFFFFFF+1=00000000 with carry=1; 0-1=FFFFFFFF with unsigned cout=0.
Test	Unsigned carry-out and borrow convention.
Result	PASS

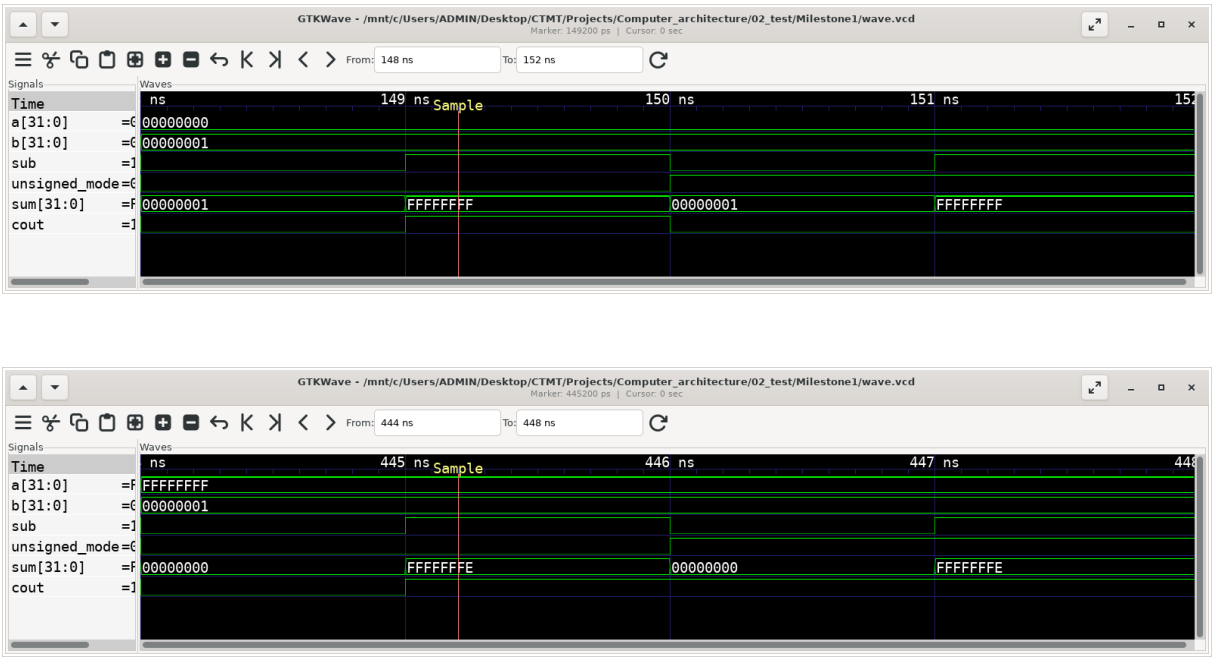


Figure WF02. Carry / borrow.

WF03 | Signed boundaries

Item	Testcase details
Input	a/b = 7FFFFFFF/1 and 80000000/1; signed mode.
Output	7FFFFFFF+1=80000000; 80000000-1=7FFFFFFF.
Test	Signed boundary wrap and status output.
Result	PASS

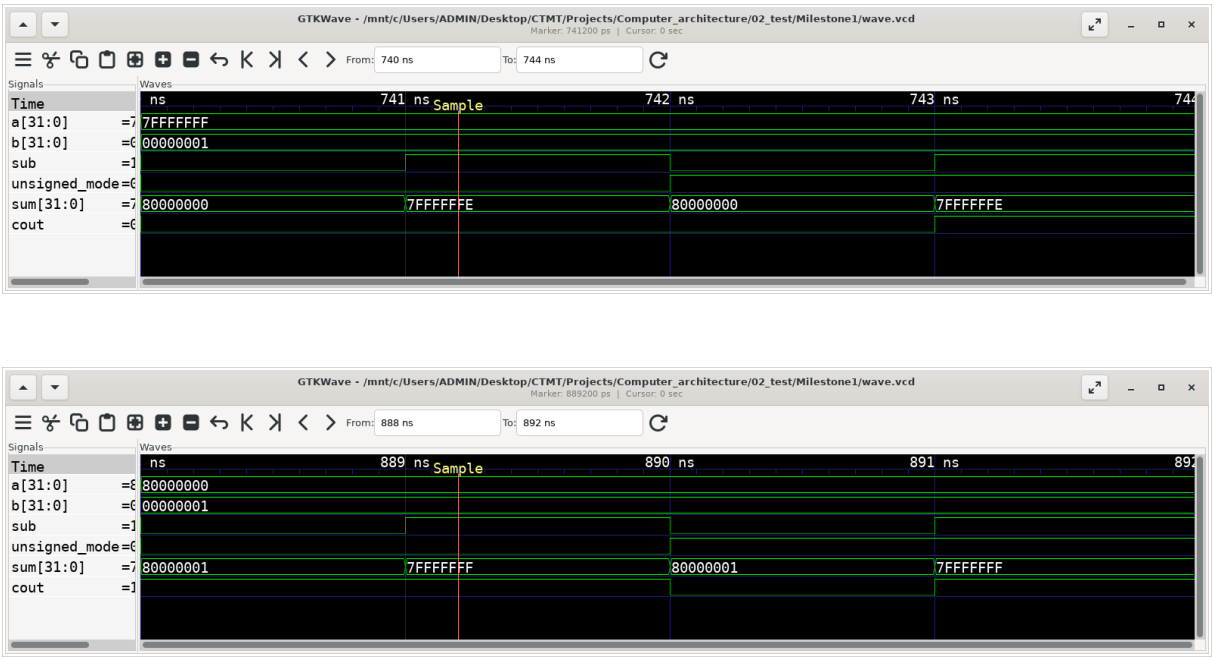


Figure WF03. Signed boundaries.

3 Magnitude Comparator

WF04 | Comparison / one-hot

Item	Testcase details
Input	a/b = 7FFFFFFF/80000000; toggle signedness; include equal operands.
Output	For 7FFFFFFF versus 80000000: signed gt=1, unsigned lt=1. Equal operands assert eq only.
Test	Ordering changes with signedness; exactly one comparison output asserted.
Result	PASS



Figure WF04. Comparison / one-hot.

4 Shifter

WF05 | Logical shifts

Item	Testcase details
Input	data=12345678; SRL/SLL; shift amounts 0,1 and 31.
Output	Shift 0 preserves 12345678; SRL1=091A2B3C; SLL1=2468ACF0; SLL31=0.
Test	Zero shift, logical direction and maximum amount.
Result	PASS

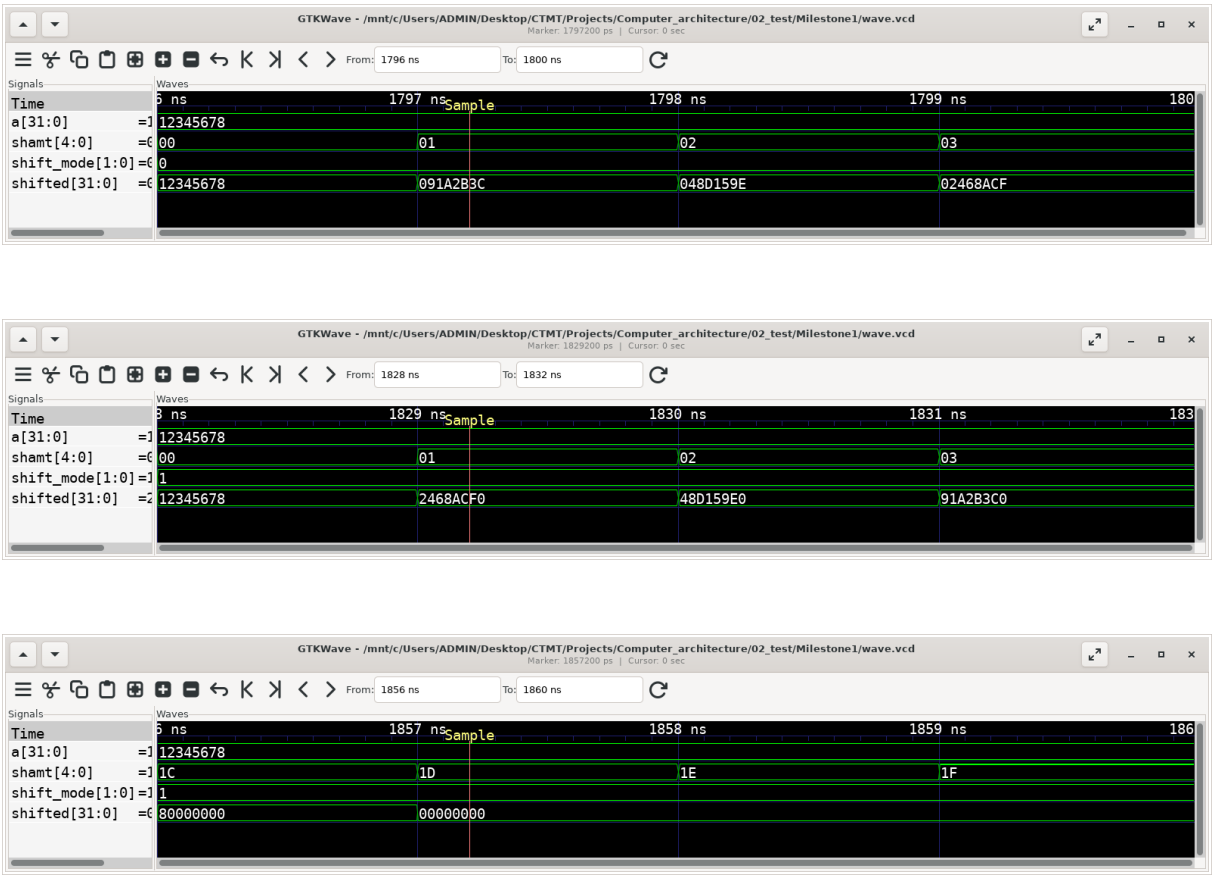


Figure WF05. Logical shifts.

WF06 | Sign fill / reserved mode

Item	Testcase details
Input	data=FEDCBA98; SRL/SRA; amounts 1,31; reserved mode=11.
Output	FEDCBA98: SRL1=7F6E5D4C; SRA1=FF6E5D4C; SRA31=FFFFFFFF. Reserved mode=0.
Test	Zero fill versus sign fill; reserved-mode safe output.
Result	PASS

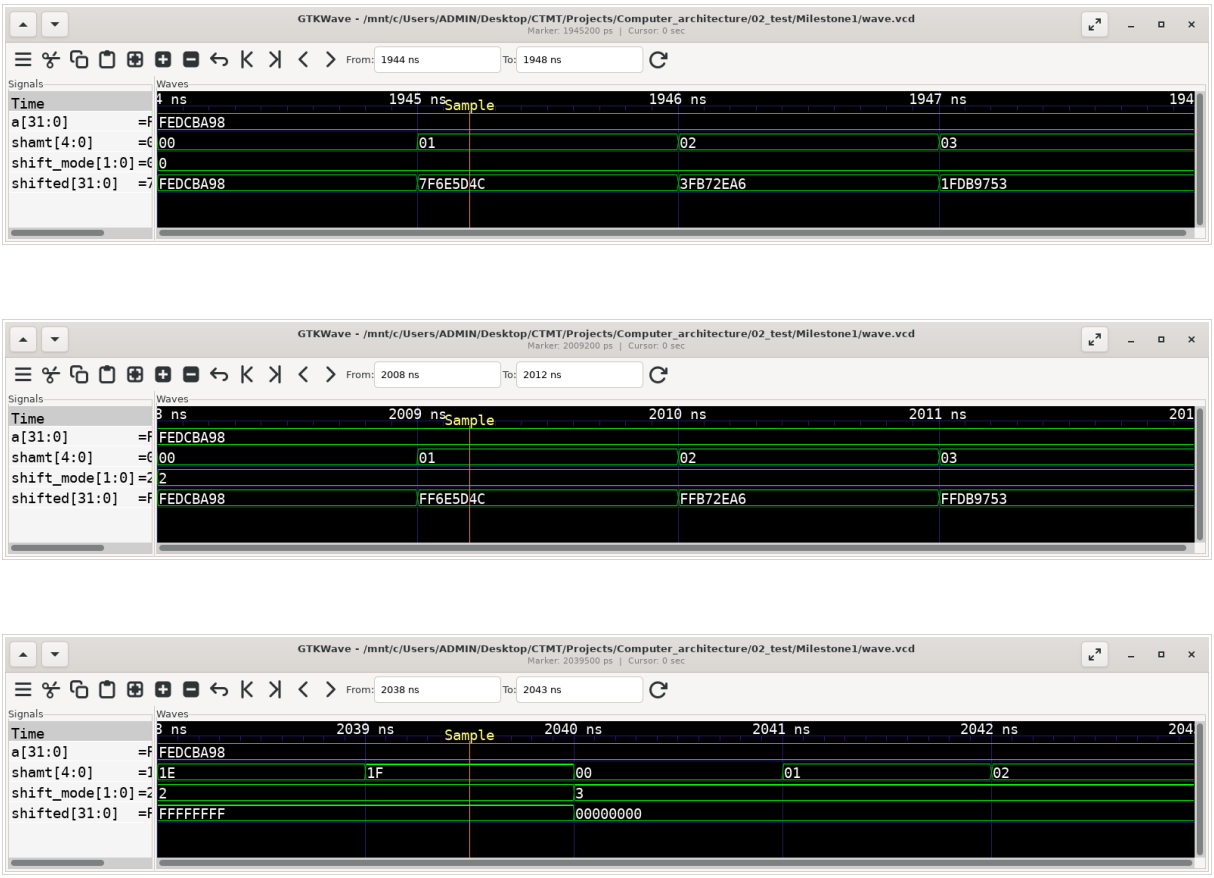


Figure WF06. Sign fill / reserved mode.

5 Arithmetic Logic Unit

WF07 | ALU selection

Item	Testcase details
Input	a=AAAAAAAA, b=55555555; sweep ALU opcodes 0-F.
Output	AAAAAAAA/55555555: XOR=FFFFFFFF, OR=FFFFFFFF, AND=0. Unused opcodes return zero.
Test	Operation selection, bitwise patterns and unused encodings.
Result	PASS

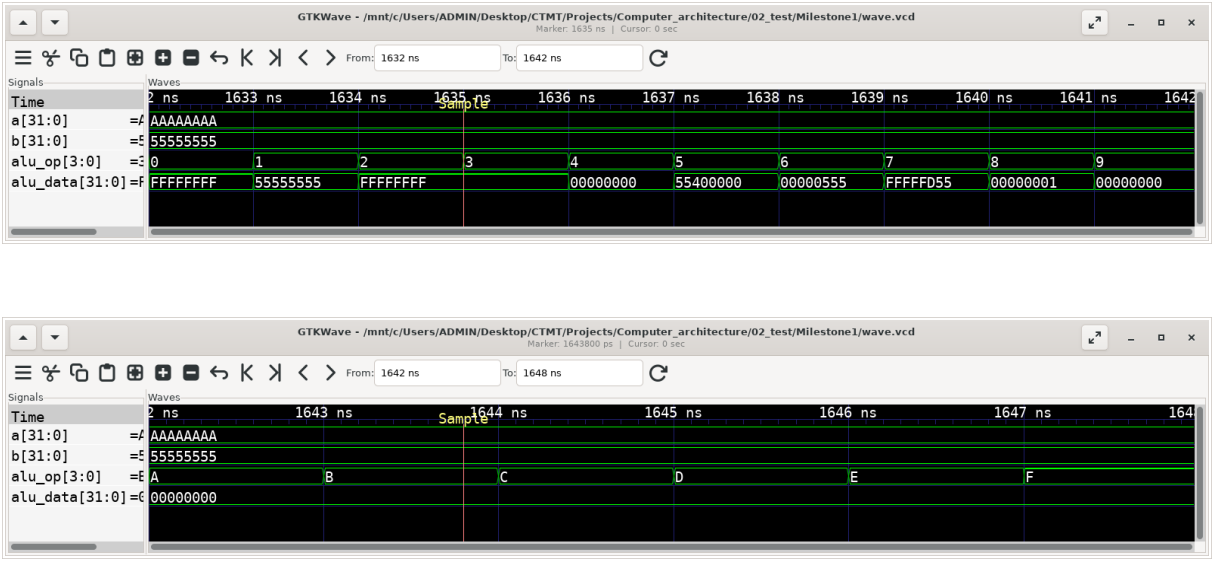


Figure WF07. ALU selection.

WF08 | Compare / shift masking

Item	Testcase details
Input	a/b=7FFFFFFF/80000000; shift b=20 or 3F hex.
Output	SLT=0 and SLTU=1 at the sign boundary. b=20 hex selects shift 0; b=3F hex selects shift 31.
Test	SLT versus SLTU; only b[4:0] selects shift amount.
Result	PASS

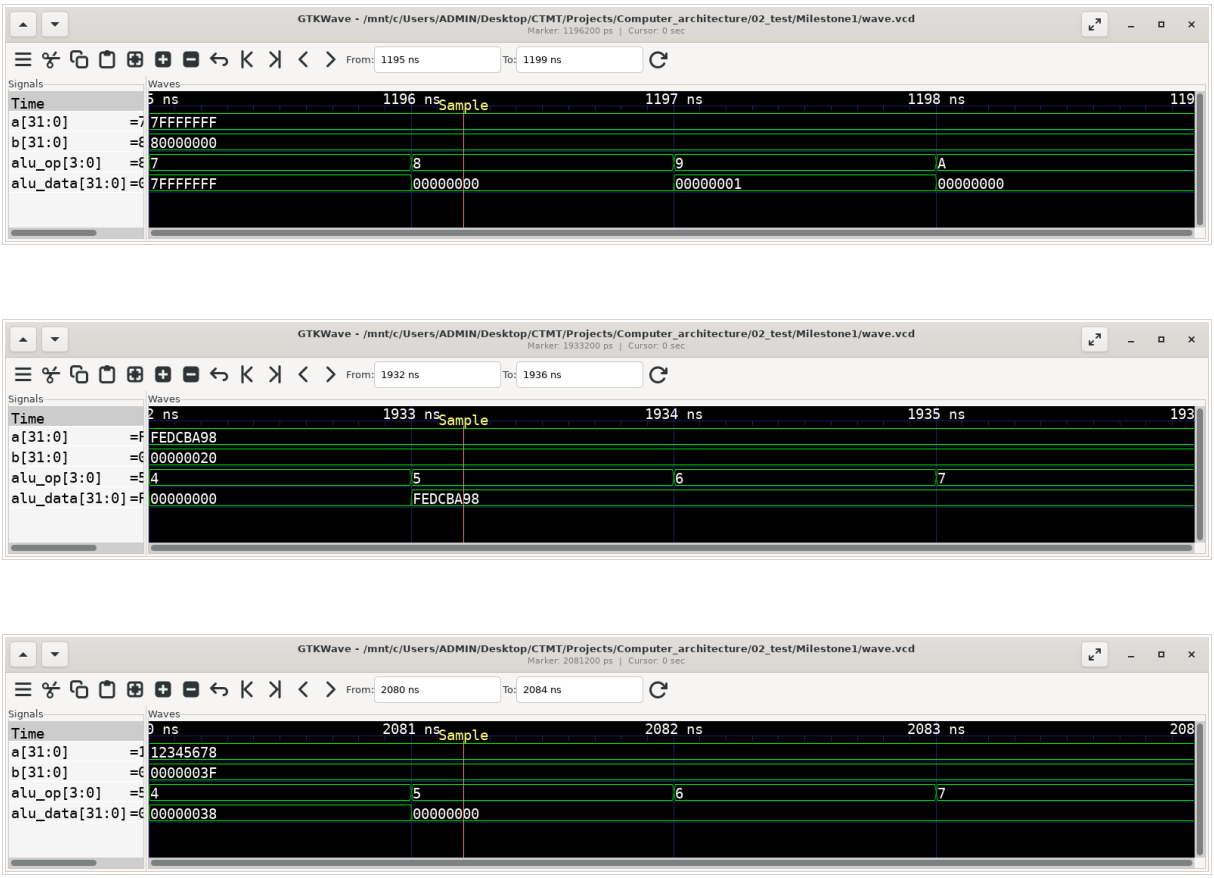


Figure WF08. Compare / shift masking.

6 Register File

WF09 | Asynchronous reset

Item	Testcase details
Input	Nonzero register contents; rstn changes 1 to 0 between rising edges.
Output	Reset asserted between rising edges clears the selected read outputs immediately; register scan returns zero.
Test	Asynchronous clearing without waiting for a clock edge.
Result	PASS

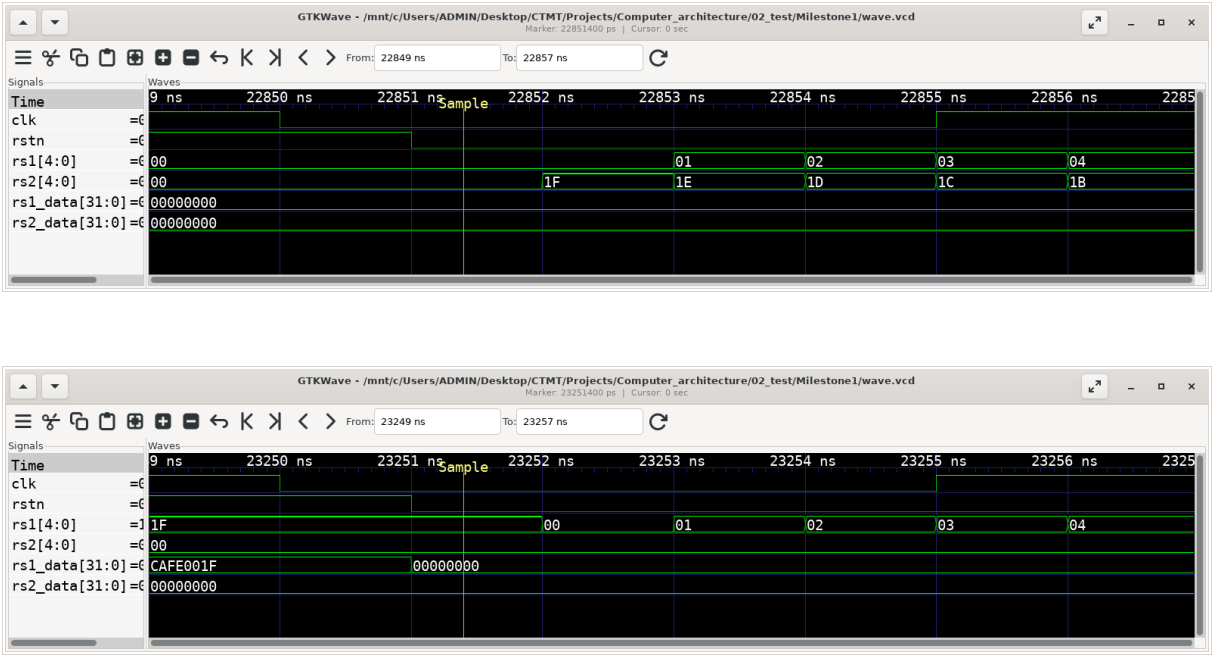


Figure WF09. Asynchronous reset.

WF10 | Register write / preservation

Item	Testcase details
Input	rd=x0-x31; write data=CAFE0000 OR index; enable then disable writes.
Output	x0 stays zero; x1 becomes CAFE0001 after the edge. Disabled write preserves x7=CAFE0007.
Test	Clock-edge update, no bypass, x0 protection and disabled-write retention.
Result	PASS

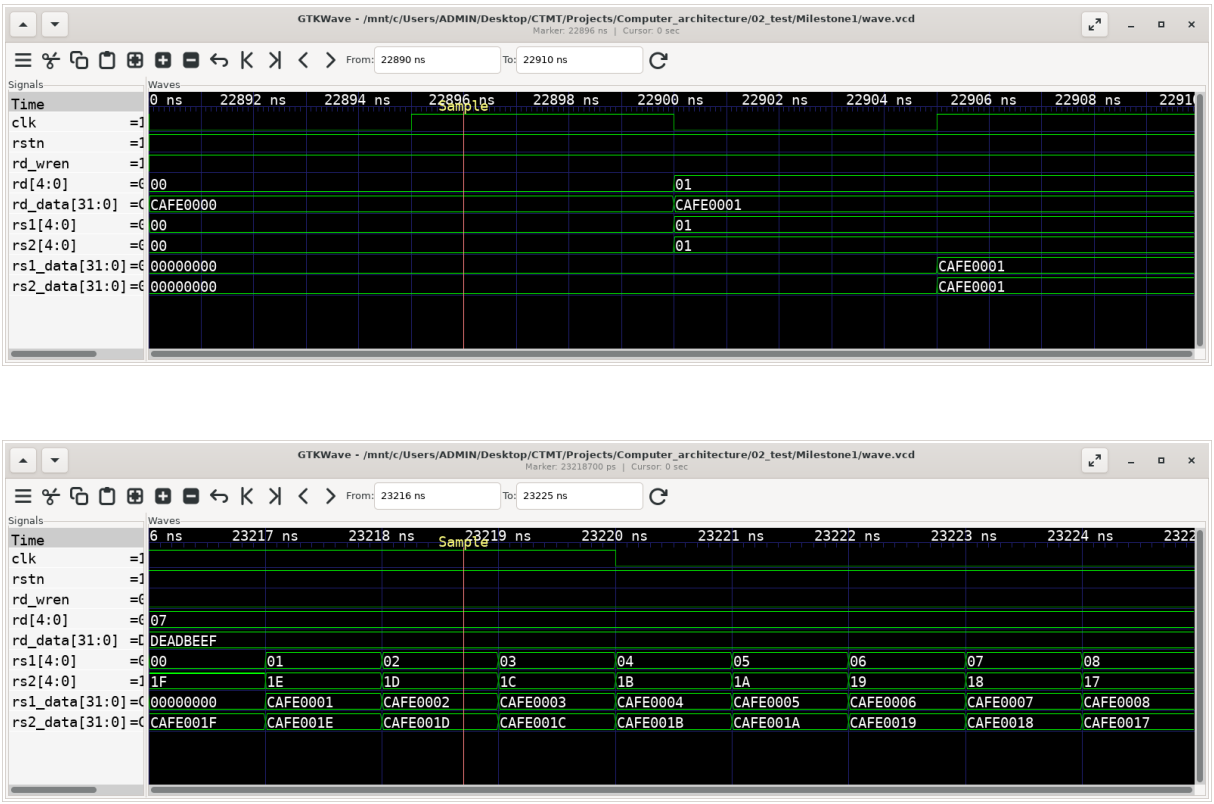


Figure WF10. Register write / preservation.

7 Asynchronous-read Memory

WF11 | Byte lanes / endianness

Item	Testcase details
Input	addr=2040-2043 and 207F; masks 0-F; full/partial word writes.
Output	Mask 0001 changes 1234AB00 to 1234ABEE. Full-word write returns 01020304; other masks preserve disabled lanes.
Test	Byte-lane preservation and little-endian data layout.
Result	PASS

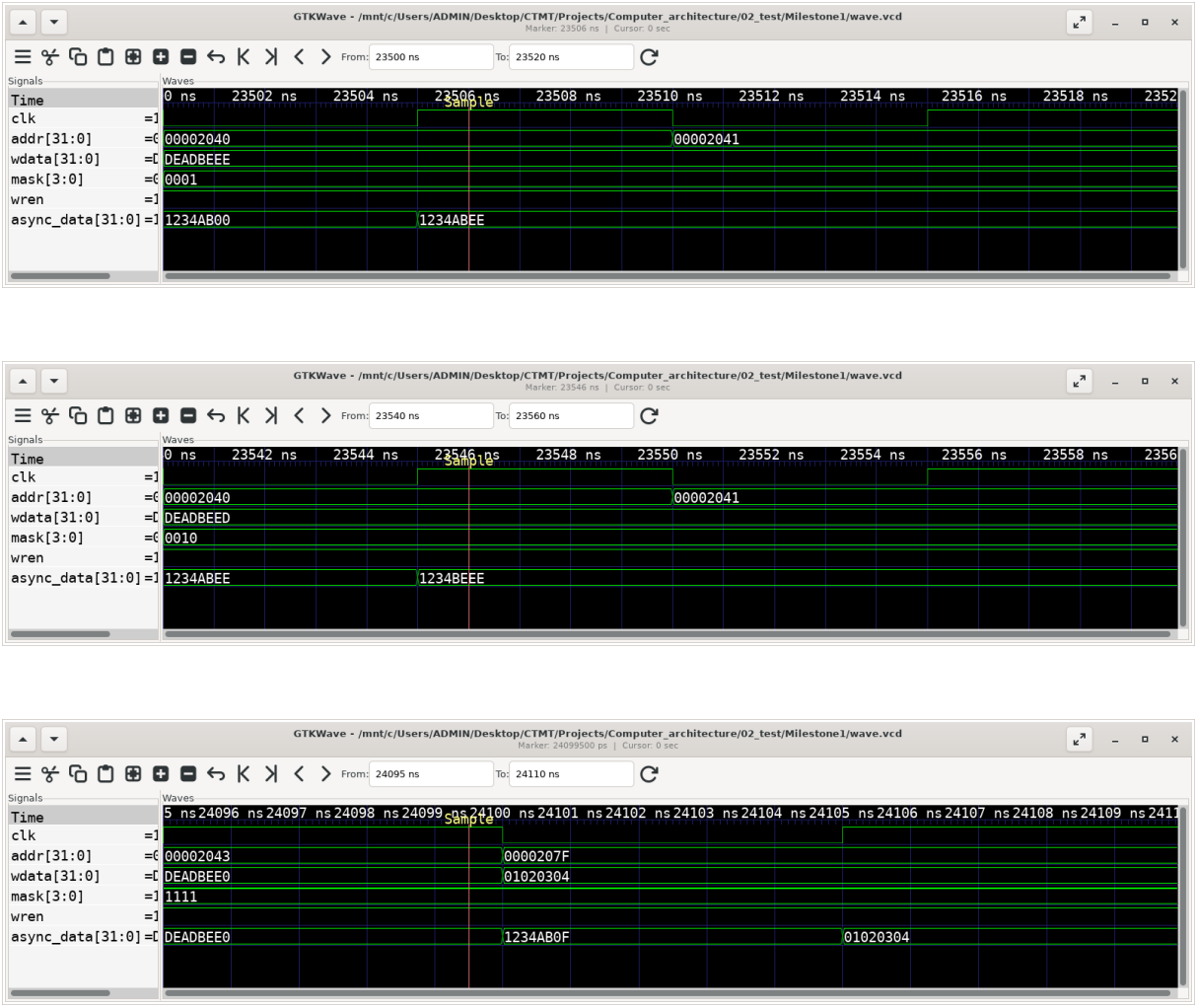


Figure WF11. Byte lanes / endianness.

WF12 | Address boundaries

Item	Testcase details
Input	Valid addr=207F; invalid addr=203F,2080,20C0,0,FFFFFFFF.
Output	Invalid reads return zero. Valid readback DEADBEE0 is unchanged after invalid writes.
Test	Upper/lower boundaries, zero invalid reads and ignored invalid writes.
Result	PASS

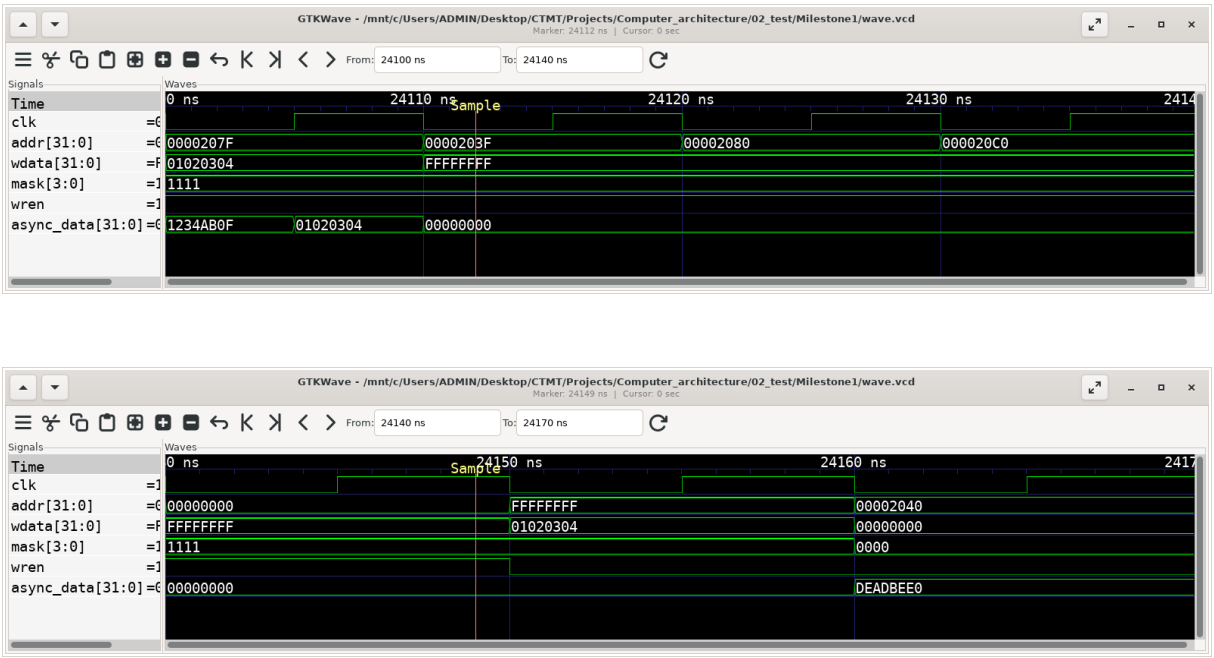


Figure WF12. Address boundaries.

8 Synchronous-read Memory (Optional)

WF13 | Registered read / hold

Item	Testcase details
Input	Read enable=0/1; change address and write enable; stored data is nonzero.
Output	Read-disable holds 01020304. Enabled read updates to 1234AB09 after the edge; enabled invalid reads return zero.
Test	Registered read timing, disabled-read hold and invalid read handling.
Result	PASS

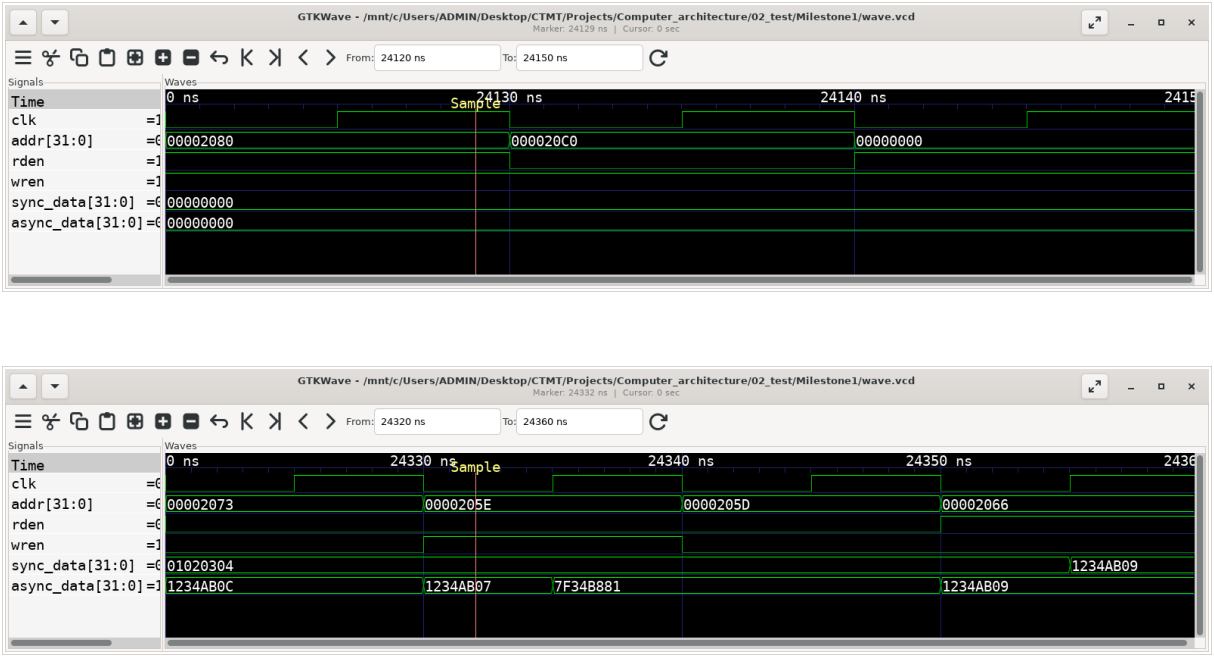


Figure WF13. Registered read / hold.

WF14 | Read-before-write

Item	Testcase details
Input	addr=2040; old=1234AB00; write=DEADBEEE; mask=0001; both enables=1.
Output	At the write edge: async=1234ABEE, sync=1234AB00. The next enabled sync read returns 1234ABEE.
Test	Simultaneous read/write uses read-before-write.
Result	PASS

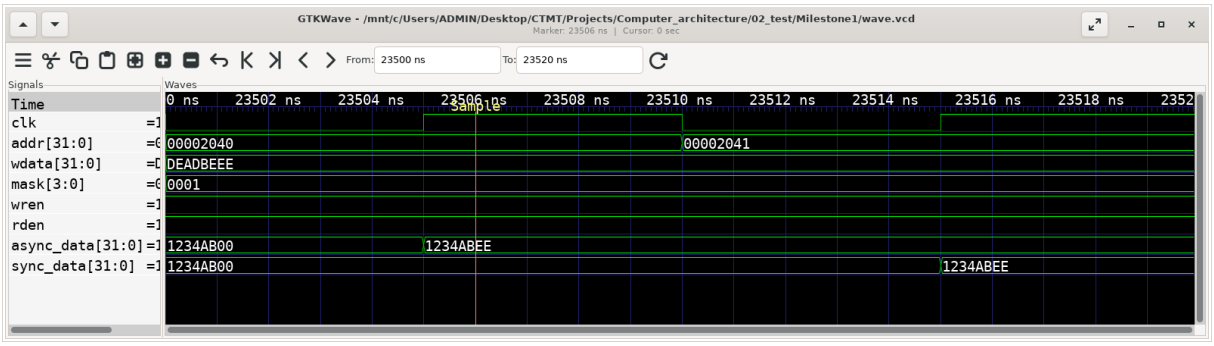
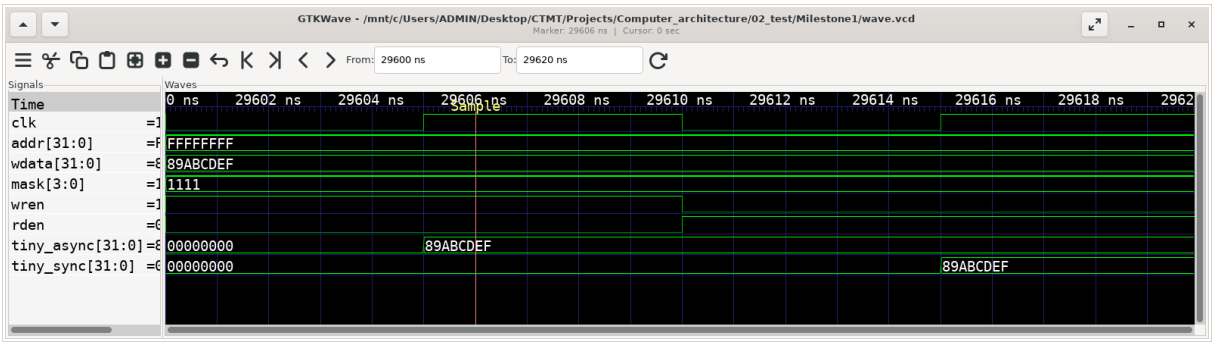


Figure WF14. Read-before-write.

WF15 | Minimum size / no wrap

Item	Testcase details
Input	One word at FFFFFFFC; write 89ABCDEF through FFFFFFFF; attempt write at 0.
Output	The top-address word retains 89ABCDEF after an invalid write at address zero; enabled invalid reads return zero.
Test	Minimum memory size and top-address no-wrap protection.
Result	PASS



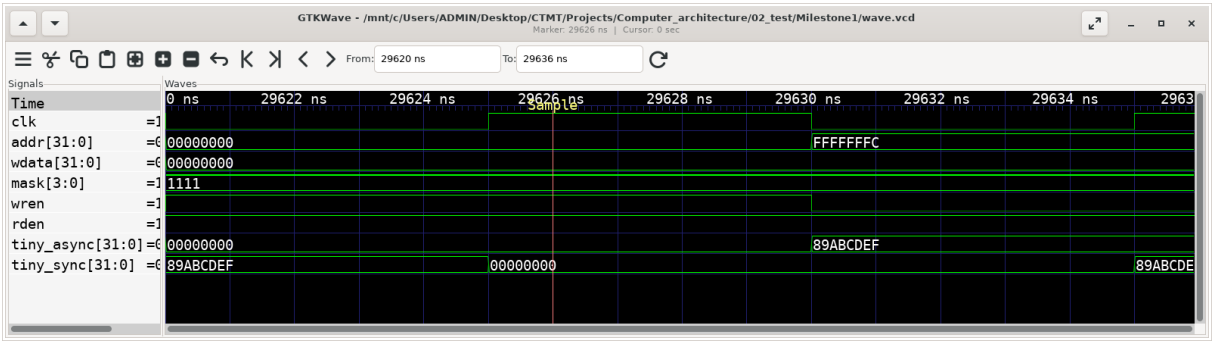


Figure WF15. Minimum size / no wrap.

9 Overall Result

Verification	Result
Standalone block tests	PASS - 46,369 checks; no mismatches.
Waveform sample checks	PASS - 15 cases, 62 checked signal values.
Waveform evidence	33 direct GTKWave screenshots.
Coverage scope	Directed and fixed-seed generated tests; not exhaustive over all 32-bit operands.